PRESS: A Player-Level Forechecking Metric

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Executive Summary

Impact. Forechecking drives zone time, suppresses breakouts, and creates turnover-based scoring chances—but individual impact is often invisible in box scores. The work is off-puck, system-dependent, and rarely a credited takeaway. To fill this gap, we introduce **PRESS** (Puck Recovery & Exit Suppression Score), a tracking-based metric that quantifies how each forechecker *increases the probability a dump-in forecheck ends in a defensive-zone turnover.*

Key insights.

- **Forecheck value is largely off-puck.** Many recoveries are forced errors (rushed rims, contested outlets) where the forechecker who created the pressure is not credited with a takeaway. PRESS captures the *pressure state* that drives those recoveries.
- **Setup vs. execution.** PRESS decomposes each forecheck into (i) **Positioning**: value created by where a player is relative to the puck and passing outlets at the start of the play, and (ii) **Execution**: value created (or lost) as the forecheck unfolds.
- **Comparable across roles and manpower.** Our approach controls for score state, home/away, and manpower (5v5, 5v4, etc.), enabling fair comparison of players in the situations they actually face.

Actionable implications.

- **Line and matchup decisions:** Identify forwards who reliably tilt dump-ins into recoveries; inform deployment and lineup choices.
- **Player development:** Use Positioning to coach setup habits (approach angle, lane denial, second-man pressure) and Execution to diagnose how effectively they sustain pressure and force errors as the forecheck unfolds.
- **Pro scouting and contracts:** Identify undervalued forecheckers whose impact is not visible in takeaways/hits.

1 Problem Framing

The dynamic, free-flowing nature of hockey creates substantial interdependence among players, making it difficult to isolate individual contributions to team outcomes. This is especially true on the forecheck: it is coached as a system, but outcomes hinge on *micro-decisions* by individuals (routes, spacing, timing). A single player a stride late can open a clean exit; a well-timed angle can eliminate the middle and force a turnover. Because turnovers are stochastic and often manifest as forced mistakes rather than direct takeaways, box-score proxies systematically undercount forecheck impact. PRESS is designed to turn that qualitative evaluation into a comparable, player-level number.

2 Data and Preprocessing

We combine Sportlogiq event data with player tracking to model each forecheck as a sequence of time-stamped snapshots. We define a forecheck as beginning when the forechecking team dumps the puck in and pressures the exiting team in the defensive zone (identified via event data). Each sequence ends at the first terminal event:

- **Recovery (Success):** forechecking team regains possession before the puck exits the zone, *or* the exiting team takes a penalty.
- **Escape (Failure):** puck exits the zone, a stoppage occurs, *or* the forechecking team takes a penalty before regaining possession.

We drop sequences where the period ends before a terminal event (unrelated to forecheck execution). The final dataset contains **28,660 forechecks** (**10,242** successes; **18,418** failures).

Snapshot representation. Rather than modeling only the terminal outcome, we compute features at each snapshot to capture how pressure evolves over the forecheck. At each frame we compute (i) **on-puck pressure**, (ii) **off-puck pressure and lane denial**, and (iii) **game-context controls**.

3 Analytical Approach

3.1 Feature construction from tracking

At each snapshot we identify the puck carrier and rank the five nearest forecheckers (F1–F5). For each forechecker we compute distance to target, radial closing speed, and approach angle; for F2–F5 we also compute pressure relative to their nearest exiting skater to capture off-puck containment. To quantify exit options, we evaluate potential passing lanes from the puck carrier to each teammate and score each lane by comparing (a) puck travel time to (b) the time for the nearest forechecker to reach an intercept zone around the lane (Appendix A). We aggregate lane features (open-lane count, middle-lane availability, nearest viable outlet distance). A full list of features is given in Appendix B.

3.2 Two-stage probability model of recoveries

We model *how likely the forechecking team is to regain possession before an exit*, and *how that probability evolves as players move.*

1. **Start model** estimates p_0, the recovery probability at the first snapshot, capturing the value of the initial configuration (spacing, lane denial, first-layer pressure).
2. **Execution model** updates that probability over snapshots by predicting recovery vs. escape at each step. Chaining these snapshot-level probabilities yields the overall probability that a forecheck ends in a recovery.

This structure cleanly separates "setting up well" from "sustaining pressure." Appendix C demonstrates the calibration of each model.

3.3 Player attribution: converting probability into individual value

For each forecheck, we define the forechecking team's "total value" as the outcome ($y \in \{0, 1\}$) minus the average success rate over all forechecks ($\bar{p}$). For example, if forechecks have an average success rate of 0.35, a forecheck that succeeds has a total value of 0.65, while a forecheck that fails has a total value of -0.35. We decompose this value into two components:

- **Positioning:** $p_0 - \bar{p}$, the probability of recovery at the start configuration relative to the average success rate.
- **Execution:** $y - p_0$, the value added (or lost) relative to the starting probability over the course of the forecheck.

To attribute team value to individual players, we allocate each component across (up to) five forechecker slots (F1–F5). Each player's share reflects how much they outperformed (or underperformed) an average "ghost" skater in that slot. Summing a player's shares across forechecks yields their per-forecheck PRESS; positive values indicate above-average contribution to forecheck success. Appendix D shows the distribution of Positioning, Execution, and total PRESS across players; Appendix E reports the top 25, and Appendix F aggregates at the team level.

4 Discussion

4.1 Results

PRESS offers an interpretable player- and team-level forechecking metric. Each player's value decomposes into Positioning (how well they set up) and Execution (how well they sustained pressure), so coaches and scouts can see not just who is valuable but why. We report player-level distributions (Appendix D), the top 25 by PRESS per forecheck (Appendix E), and team-level aggregates (Appendix F). The top 25 span forwards and defensemen, with some excelling at setup and others at sustaining pressure.

4.2 Implications for Teams

PRESS is intended to complement video and coaching judgement, not replace it. Its value is attaching a *probabilistic* impact to the habits coaches already teach:

- **Scouting:** surface forecheck drivers whose impact is hidden in traditional stats.

- **Usage:** choose personnel for dump-in heavy game plans, protect leads by prioritizing strong lane denial, or chase games with high-execution forecheckers.
- **Development:** diagnose whether a player's issue is *setting up well* (Positioning) or *sustaining pressure* as the play unfolds (Execution), enabling targeted coaching.

4.3 Assumptions and Limitations

- **System dependence:** forechecking is coordinated; PRESS controls for context and uses counterfactual replacements, but cannot fully disentangle coaching instructions from player decisions.
- **Tracking proxies:** lane-blocking uses an intercept-time proxy (Appendix A); different assumptions about pass speed or reaction time change absolute values (rankings are more stable).
- **Tracking dropout:** when video tracking is missing for a forechecker or snapshot, we impute slot features; sequences without sufficient tracking may be excluded. This can bias estimates if dropout correlates with play type or venue.

4.4 Reproducibility

Code and instructions to reproduce our analysis are available at github.com/WhartonSABI/halo2026.

Appendix

A Lane-Blocking Proxy Details

We score each potential outlet pass from puck carrier to teammate by comparing puck travel time along the lane to the time for the nearest forechecker to reach an intercept zone around that lane. We treat a lane as blocked when the defender arrives first (negative margin). Implementation constants (intercept radius, assumed pass speed, reaction time) are listed in `02_features.py`.

B Full Feature Set

Carrier state Puck carrier position (x, y) and speed at each snapshot.

On-puck pressure For forecheckers 1–5 (F1–F5): distance to carrier (r), radial closing speed (v_r), approach angle ($\sin\theta$, $\cos\theta$), lane-block severity (overall and center).

Off-puck and lane denial For forecheckers 2–5 (F2–F5): distance and closing speed to nearest opponent along with lane aggregates: outlet candidate count, unblocked outlet count, center-open indicator, minimum unblocked outlet distance.

Game-context Manpower state (e.g., 5v5), pressing-team home indicator, score-difference bin (trailing/tied/leading), puck location at forecheck start (x, y).

Time Time since forecheck start (seconds), plus some simple transformations in the hazard model (log, sqrt, spline basis).

C Calibration Plots

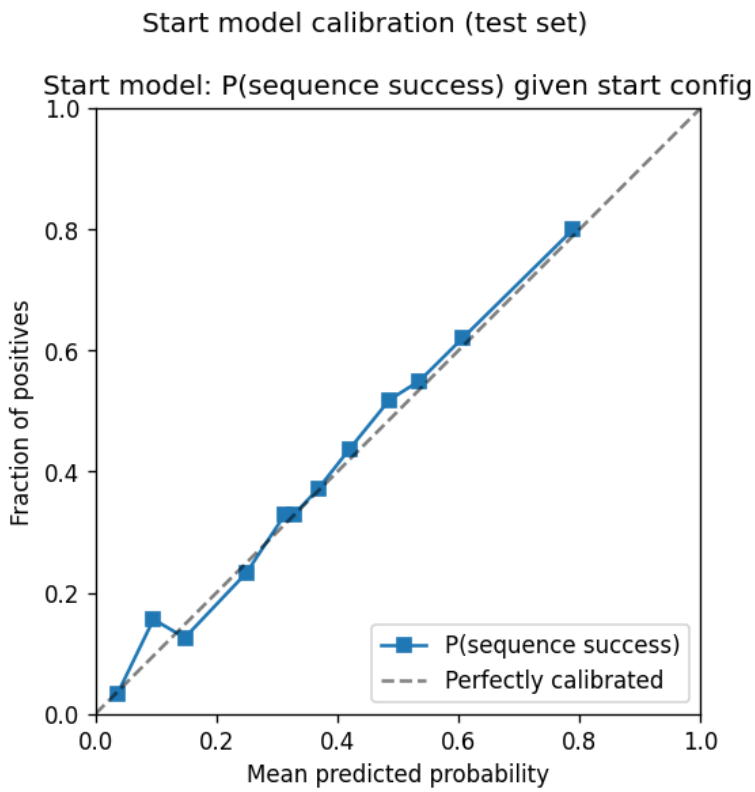

Figure 1: Start model p_0: out-of-sample reliability diagram.

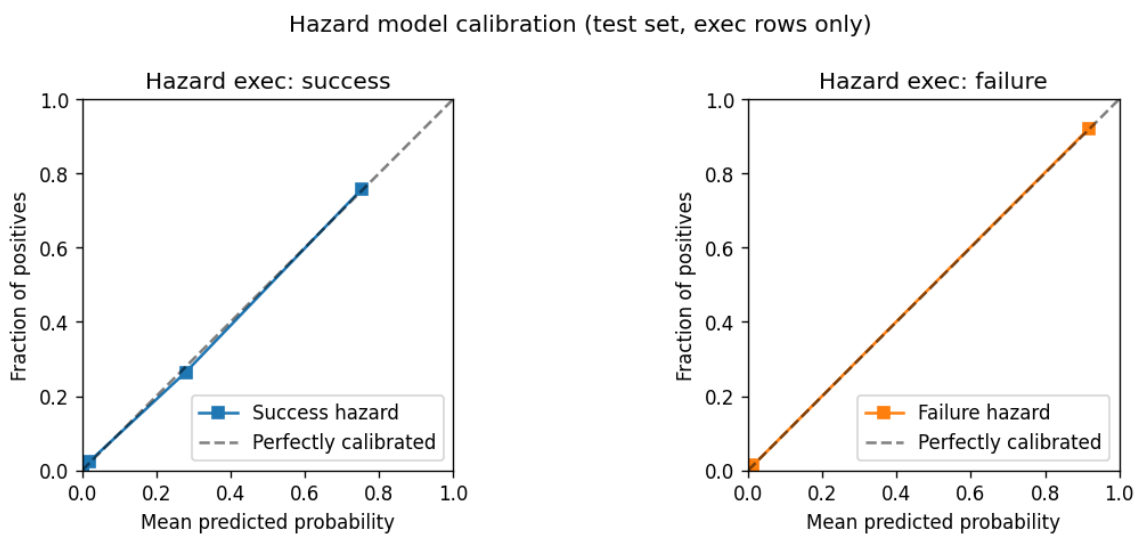

Figure 2: Execution hazards: out-of-sample reliability diagrams.

D Contribution Distributions

Figure 3 shows the distribution of Positioning, Execution, and total PRESS across all players.

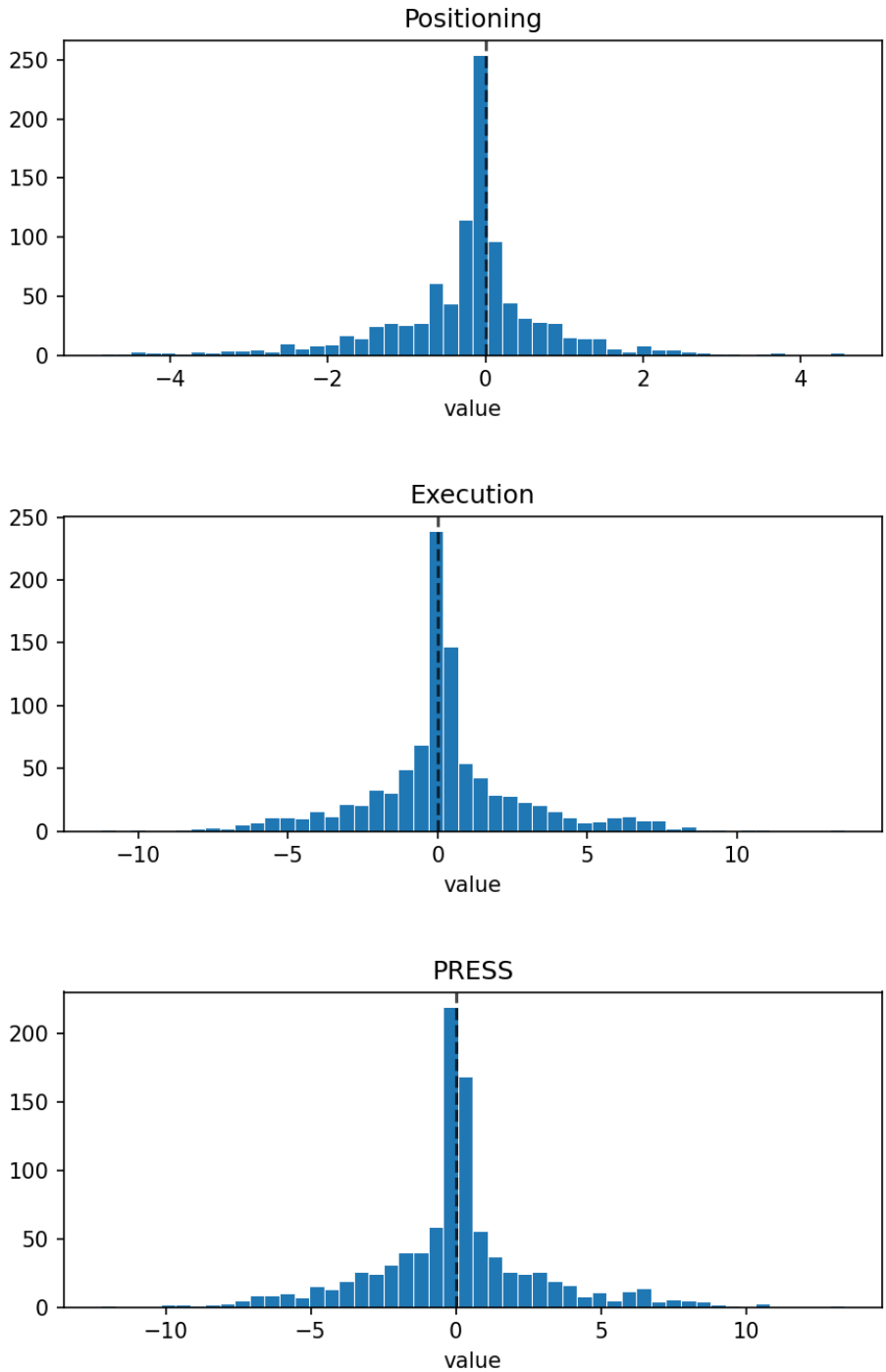

Figure 3: Player-level distributions of Positioning (top), Execution (middle), and PRESS (bottom).

E Player Leaderboard

Table 1 reports the top 25 forecheckers by PRESS per forecheck (minimum 10 forechecks for inclusion). Pos and Exec decompose each player's contribution into positioning (start-configuration value) and execution (how the play unfolded). All values are per forecheck.

Table 1: Top 25 forecheckers by PRESS per forecheck.

Rank	Player	Position	n	Pos	Exec	PRESS
1	Bowlby, Henry	LW	15	0.030	0.169	0.200
2	Lapierre, Hendrix	C	42	0.022	0.094	0.116
3	Firstov, Vladislav	F	27	0.026	0.082	0.108
4	Robertson, Nick	LW	16	0.019	0.069	0.089
5	Kiviranta, Joel	LW	21	0.024	0.061	0.085
6	Carpenter, Ryan	C	33	0.019	0.063	0.082
7	Zary, Connor	C	11	0.026	0.052	0.078
8	Alexandrov, Nikita	C	15	−0.020	0.097	0.077
9	Caamano, Nicholas	RW	184	−0.001	0.074	0.073
10	Hamblin, James	C	24	0.016	0.057	0.072
11	Armia, Joel	RW	10	−0.001	0.069	0.068
12	Sandelin, Ryan	F	13	0.035	0.033	0.068
13	Anderson, Joey	RW	35	0.017	0.049	0.066
14	Carroll, Joe	LW	130	0.005	0.058	0.063
15	Crozier, Max	D	104	0.001	0.060	0.061
16	Luneau, Tristan	D	24	0.008	0.051	0.059
17	Anderson-Dolan, Jaret	C	24	0.028	0.031	0.059
18	Ivan, Ivan	F	14	0.031	0.026	0.058
19	Kniazev, Artemi	D	66	0.019	0.038	0.057
20	Beaudin, Nicolas	D	29	0.005	0.050	0.056
21	Marino, Kyle	RW	16	0.000	0.056	0.056
22	Fortier, Gabriel	C	142	0.001	0.053	0.055
23	McDonough, Aidan	LW	193	0.011	0.043	0.054
24	Sutter, Brett	LW	135	0.010	0.044	0.054
25	Henman, Luke	F	73	0.022	0.030	0.052

F Team-Level PRESS

Figure 4 aggregates PRESS at the team level.

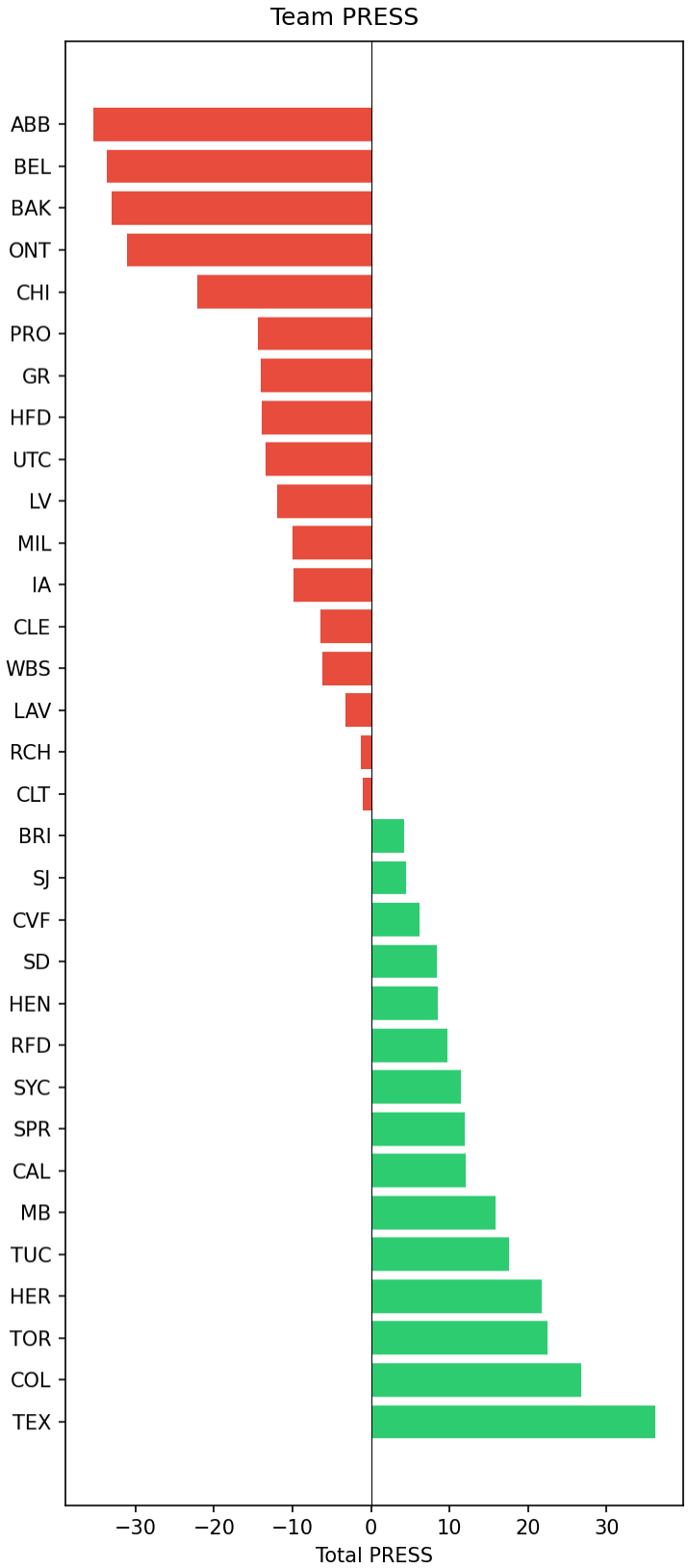

Figure 4: Team-level total PRESS.